

Narendra Singh

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Education

Indian Institute of Technology Kanpur

Bachelor of Technology in Computer Science & Engineering

Nov 2021 – May 2025

CGPA: 8.5/10

Scholastic Achievements

- Secured **All India Rank 444** (Top 0.3%) in the JEE-Advanced (2021) and **All India Rank 1126** in JEE-Mains (2021)
- Awarded Academic Excellence Award (2021-22) by IIT Kanpur for ranking in the **Top 10%** of the batch
- Received **Vahani Scholarship 2021**, awarded to 35 meritorious students across India by the **Vahani Scholarship trust**

Experience

Software Developer (SDE - 1)

July 2025 – Apr 2026

InMobi | Ad-Serving Team

Bengaluru, India

- Optimized **Ad-Serving Microservices** by performing root cause analysis, reducing system-wide error volume by **80%**
- Migrated a Scala-based Ad-Classifer service from OpenAI to Gemini, implementing GKE cluster-level authorization
- Engineered a resilient **fallback mechanism** using historical data trends to classify creative assets, achieving **100% classification coverage** and eliminating critical data gaps
- Integrated **Prometheus Pushgateway** to capture and emit quantitative metrics from short-lived jobs for **observability**
- Contributed to A/B testing framework for bid-request signals for quantitative assessment of data attributes across DSPs
- Investigated and resolved production incidents alongside on-call engineers, significantly **minimizing system downtime**

Research and Development Internship

May 2024 – Jul 2024

*Samsung R & D Institute India | Received a **Pre-Placement Offer** for exceptional performance*

Noida, India

- Implemented a real-time hand gesture recognition model (PairNet) for smartwatches, achieving a **92.36%** hit rate
- Architected a statistical preprocessing pipeline utilizing **PCA** & Feature Selection, reducing training data size by **40%**
- Improved model accuracy by 3% via custom skip connections, contributing to a pre-commercial digital health feature

Projects

Static Hash Mapping | *Randomized Algorithms Course Project* | *Prof. Surender Baswana*

Jan 2025 – Apr 2025

- Constructed a Las Vegas randomized algorithm utilizing perfect hashing to guarantee $O(1)$ worst-case lookup query time
- Leveraged universal and perfect hashing to achieve a strictly bounded $< 6n$ space footprint, providing a significant performance uplift over binary search methods through $O(n)$ preprocessing and $O(1)$ worst-case retrieval speed

Python Compiler | *Compiler Course Project* | *Prof. Swarnendu Biswas* | [Github](#)

Jan 2024 – May 2024

- Architected a scanner-parser process for generating and visualizing Abstract Syntax Trees (ASTs) from input Python
- Extended by implementing a symbol table data structure, conducting limited error checking during semantic analysis
- Engineered a translator to convert the generated intermediate three-address code into x86_64 assembly instructions

E-Vaidya | *Software Development Course Project* | *Prof. Indranil Saha* | [Github](#)

Jan 2023 – May 2023

- Developed Automated appointment procedures at the Institute Health Centre with a web-based application
- Used bcrypt, a NodeJS package to store the hashed password and JWT Tokens to provide authorization for each user

Relevant Courses

Systems & Architecture: Operating Systems, Parallel Computing, Computer Networks

Theory & Algorithms: Data Structures and Algorithms, Algorithm Design

Data & AI: Principles of Database Systems, Introduction to Machine Learning, Probability for CS

Skills

Languages: C/C++, Python, SQL, Java, Scala | **Tools:** Git, Google Cloud Platform, Linux, Airflow

Specialization: Backend Architecture, Distributed Systems, Machine Learning, Data Structures & Algorithms, OOP

Position of Responsibility

Leader, Archery Society | *Games and Sports Council, IIT Kanpur*

May 2023 – Apr 2024

- Spearheaded a team of 15 secretaries in revamping the campus archery culture, resulting in increased participation
- Secured vital approvals and drafted MOUs for partnerships & events, collaborating with leading training academies